Akoizo Original Practice - 2027 Rotation Form 4

Division C | Circuit Lab | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) XOR(1,1) equals…

Unit: Digital logic and combinational circuits.
Worked-case data: For A=1, B=1, find half-adder sum and carry.

A. 0

B. 1

C. 2

D. Undefined

Answer: _____

2. (2 points) Two 100 Ω resistors in series give…

Unit: Diagrams, nodes, series, parallel, and Ohm's law.
Worked-case data: Find Req for 100 Ω and 300 Ω in parallel on 6 V.

A. 200 Ω

B. 50 Ω

C. 100 Ω

D. 10000 Ω

Answer: _____

3. (2 points) An ohmmeter generally requires…

Unit: Breadboards, multimeters, and a first virtual build.
Worked-case data: A known 100 Ω resistor has 2 V across it; an unknown in series has 4 V. Find the unknown.

A. A live mains circuit

B. Maximum possible voltage

C. No instrument instructions

D. A de-energized suitable circuit

Answer: _____

4. (2 points) A multiplexer…

Unit: Digital logic and combinational circuits.
Worked-case data: For A=1, B=1, find half-adder sum and carry.

A. Always adds inputs

B. Stores a clocked bit by definition

C. Measures resistance

D. Selects an input

Answer: _____

5. (4 points) Why trace nodes first?

Unit: Diagrams, nodes, series, parallel, and Ohm's law.
Worked-case data: Find Req for 100 Ω and 300 Ω in parallel on 6 V.

Answer: _____
_____
_____

6. (4 points) Why check rail continuity?

Unit: Breadboards, multimeters, and a first virtual build.
Worked-case data: A known 100 Ω resistor has 2 V across it; an unknown in series has 4 V. Find the unknown.

Answer: _____
_____
_____

7. (4 points) Why test all input combinations?

Unit: Digital logic and combinational circuits.
Worked-case data: For A=1, B=1, find half-adder sum and carry.

Answer: _____
_____
_____

8. (4 points) Why does divider loading matter?

Unit: Diagrams, nodes, series, parallel, and Ohm's law.
Worked-case data: Find Req for 100 Ω and 300 Ω in parallel on 6 V.

Answer: _____
_____
_____

9. (4 points) Why isolate an unknown for resistance measurement?

Unit: Breadboards, multimeters, and a first virtual build.
Worked-case data: A known 100 Ω resistor has 2 V across it; an unknown in series has 4 V. Find the unknown.

Answer: _____
_____
_____

10. (4 points) What does a Boolean model omit?

Unit: Digital logic and combinational circuits.
Worked-case data: For A=1, B=1, find half-adder sum and carry.

Answer: _____
_____
_____

11. (2 points) A 10 V ideal battery drives a 5 ohm resistor. Find current.

Answer: _____
_____
_____

12. (2 points) A series pair has resistances 6 and 8 ohms. Find equivalent resistance.

Answer: _____
_____
_____

13. (2 points) Two identical 12 ohm resistors are placed in parallel. Find equivalent resistance.

Answer: _____
_____
_____

14. (2 points) An ideal resistor carries 2 A at 10 V. Find power.

Answer: _____
_____
_____

15. (2 points) A node has incoming currents 7 A and 2 A and one outgoing current. Find the outgoing current.

Answer: _____
_____
_____

16. (2 points) Explain why an ideal voltmeter is connected in parallel rather than used as a series current path.

Answer: _____
_____
_____

17. (2 points) Give the output of an AND gate for inputs 1 and 0, and explain.

Answer: _____
_____
_____

18. (2 points) A silicon diode is reverse biased below breakdown. Contrast its ideal current with a forward-biased conducting diode.

Answer: _____
_____
_____

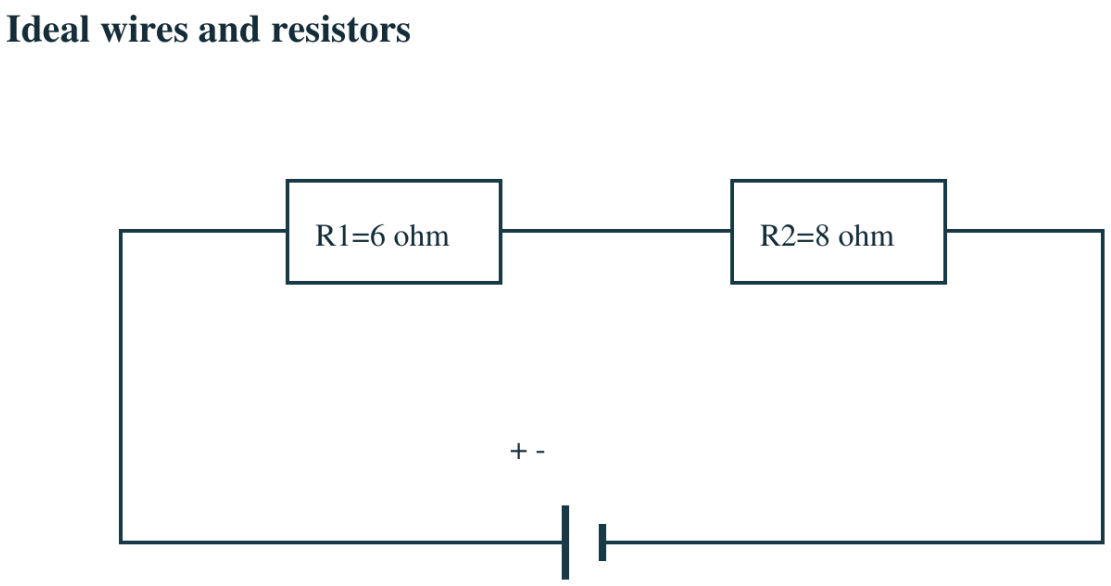
19. (2 points) A closed DC loop has a 9 V source and a 4 V resistor drop. Find the remaining series resistor drop and state the governing law.

Answer: _____
_____
_____

20. (2 points) Two identical LEDs must be equally bright in a low-voltage teaching circuit. Explain why their branch currents and current-limiting arrangements matter.

Answer: _____
_____
_____

21. (2 points) Use Figure F1. Calculate equivalent resistance.



F1. Series resistor circuit (original synthetic teaching figure)

Answer: _____
_____
_____

22. (2 points) Use Figure F1. Calculate loop current.

Refer to F1. Series resistor circuit (original synthetic teaching figure) shown with the first question in this figure group.

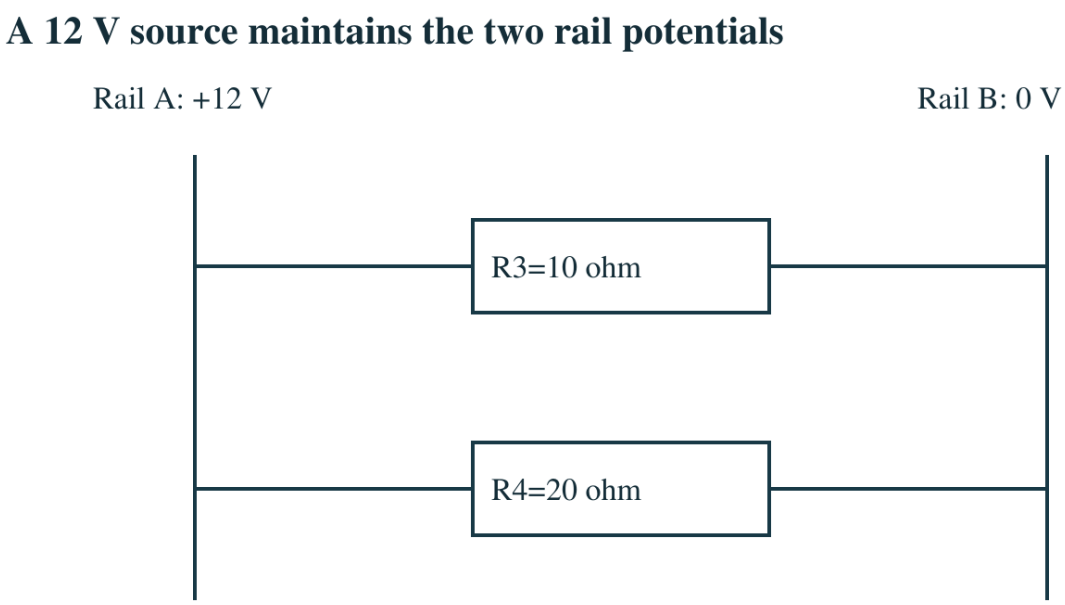
Answer: _____
_____
_____

23. (2 points) Use Figure F1. Calculate the potential difference across R1.

Refer to F1. Series resistor circuit (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

24. (2 points) Use Figure F2. Find current through R3.



F2. Two parallel branches (original synthetic teaching figure)

Answer: _____
_____
_____

25. (2 points) Use Figure F2. Find current through R4.

Refer to F2. Two parallel branches (original synthetic teaching figure) shown with the first question in this figure group.

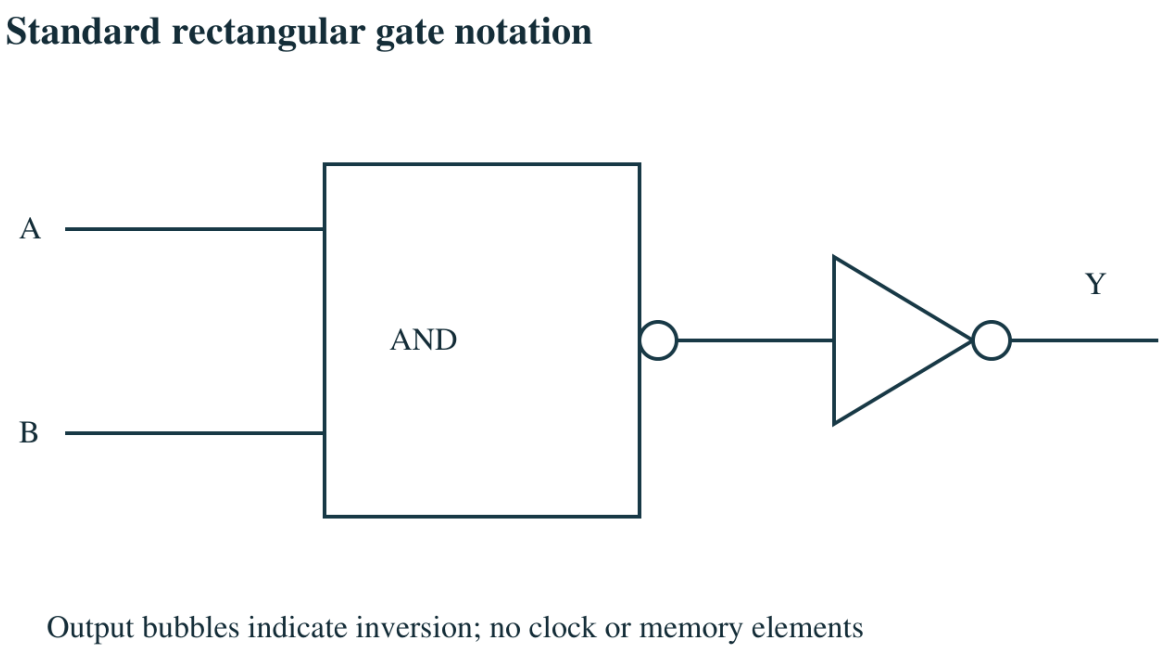
Answer: _____
_____
_____

26. (2 points) Use Figure F2. Find equivalent resistance between the rails.

Refer to F2. Two parallel branches (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

27. (2 points) Use Figure F3. Find Y for A=1 and B=0.



F3. Combinational logic (original synthetic teaching figure)

Answer: _____
_____
_____

28. (2 points) Use Figure F3. Find Y for A=1 and B=1.

Refer to F3. Combinational logic (original synthetic teaching figure) shown with the first question in this figure group.

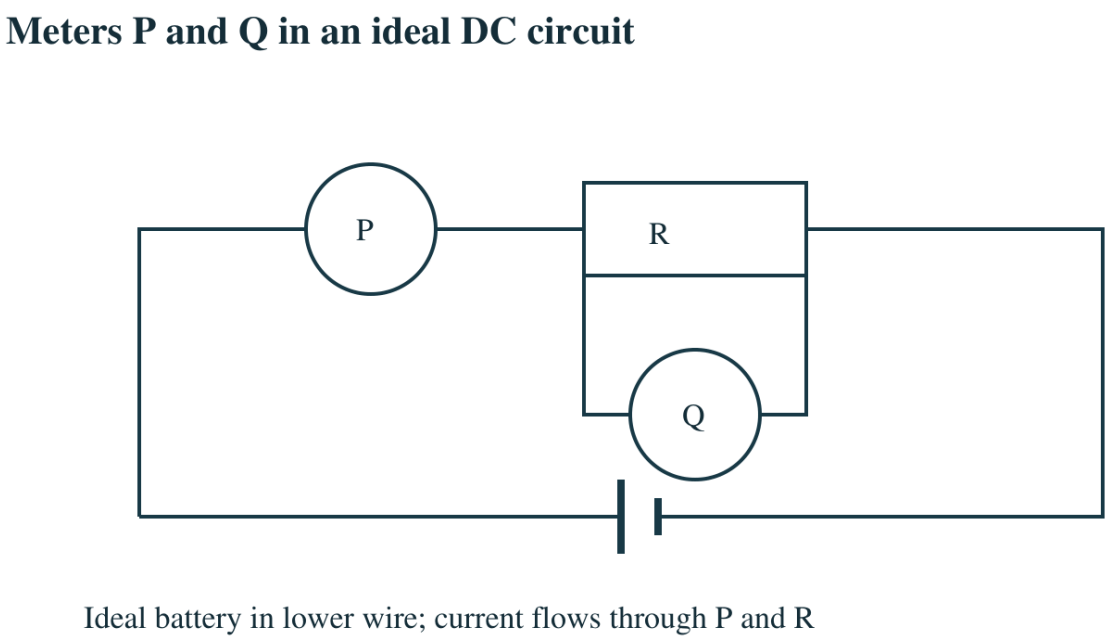
Answer: _____
_____
_____

29. (2 points) Use Figure F3. Name the single gate equivalent to the complete diagram and justify.

Refer to F3. Combinational logic (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

30. (2 points) Use Figure F4. Which meter position is appropriate for a voltmeter? Explain using the wiring.



F4. Measurement connections (original synthetic teaching figure)

Answer: _____
_____
_____

31. (2 points) Use Figure F4. Which meter position is appropriate for an ammeter? Explain.

Refer to F4. Measurement connections (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

32. (2 points) Use Figure F4. Describe the desired input resistance of the meter at Q and its purpose.

Refer to F4. Measurement connections (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____